

CLOUD COMPUTING COURSE

Lecture18-Cloud Storage (1)

ABSTRACT
Lecture18 Introduces storage mainstream types

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Storage Virtualization Definition

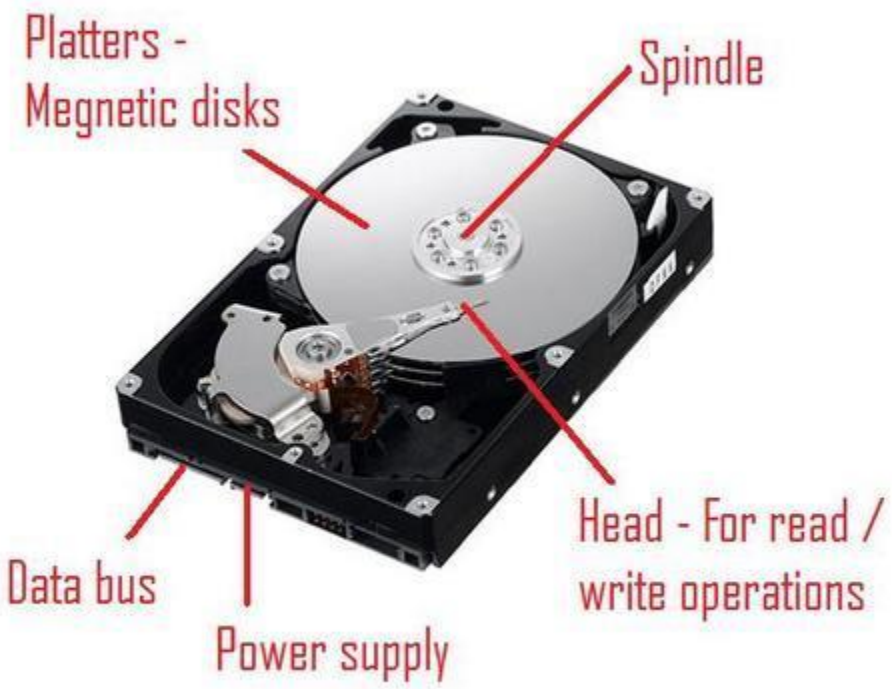
Storage virtualization is the process of grouping and sharing the physical storage from multiple network storage devices so that it looks like a single storage device, which simply managed from a central console.

When you buy a server or storage array these days, you often have the choice between different kinds of hard drives:

Mainstream Physical Disk Types

Type1: HDD

HDDs (Hard Disk Drive): are considered as an irreplaceable and important storage device that uses a **mechanical arm** with a **read/write head** to move around and read information from the right location on a **storage platter**.



HDD Types:



Currently, SATA and SAS hard drives are commonly used and **distinguished by interface** type.

1. **PATA (Parallel advanced technology attachment):** this transfers data in parallel mode and it is not used any more.

2. **SATA hard drive (Serial advanced technology attachment):** it replaced all PATA interfaces, SATA transfers data in serial mode. One of its distinctive features is that SATA is faster than PATA and have a very high capacity.



SATA HDD

3. **SAS**

   Serial attached SCSI (SAS), that is, small computer system interface. Similar to SATA, SAS uses serial transfer to achieve higher performance. Compared with a SATA hard drive, a SAS hard drive has a higher read/write speed, higher price, and **smaller c**apacity.



SAS HDD

In reliability, SAS disks are an order of magnitude safer than either NL-SAS or SATA disks. This metric is measured in *bit error rate* (BER), or how often bit errors may occur on the media.

4. NL-SAS

Near Line SAS (NL-SAS) is a type of hard drive between SATA and SAS. Therefore, an NL-SAS hard drive is developed, which consists of SAS interfaces and SATA platters (head, media, and rotational speed).



NL_SAS HDD

NL-SAS is slower than SAS.

NL-SAS is faster than SATA.

NL-SAS has the capacity and price of the SATA hard drive and the reliability of the SAS hard drive.

Type2: SSD (Solid-State Drives)

- SSDs offer up to 100 time's greater throughput than traditional hard drives, which means **faster boot ups and better overall system performance**.
- Unlike SATA and SAS hard drives, SSDs **have no moving parts** making them the most reliable and efficient option.
- The lack of moving parts also means a **reduced risk of failure** and an **increase** in **power efficiency** compared to traditional hard drives.
- SSDs, as storage media with **higher read/write speeds** than traditional HDDs, have received widespread attention.
- Traditional HDDs are disk drives and data is stored in disk sectors. The common storage medium of an **SSD is flash memory**.
- Because of the performance, speed, and reliability benefits, solid-state drives are generally **more expensive** than SAS and SATA hard drives.
- SSDs are ideal **for high frequency immediate transactional data like database, CRM or bank transactions.**
- SSDs are one of the most essential storage devices in the **ultra-thin laptop** and tablet fields.
- In addition, SSDs have distinctive features **such as shock resistance, small size, no noise, and low cooling requirements.**
- An SSD consists of **a flash controller and memory chips**. The flash controller controls the coordination of the data read/write process and the memory chips are responsible for storing data.
- A most common type is to use **a flash memory chip as the storage medium**, and the other type is to use **a dynamic random access memory** (DRAM) chip as the storage medium.



SSD

Flash-based SSDs

The most common SSDs use a flash memory chip as the storage medium. Flash memory chips can be manufactured into various electronic products, such as SSDs, memory cards, and USB drives. These devices have **small size and easy to use.**

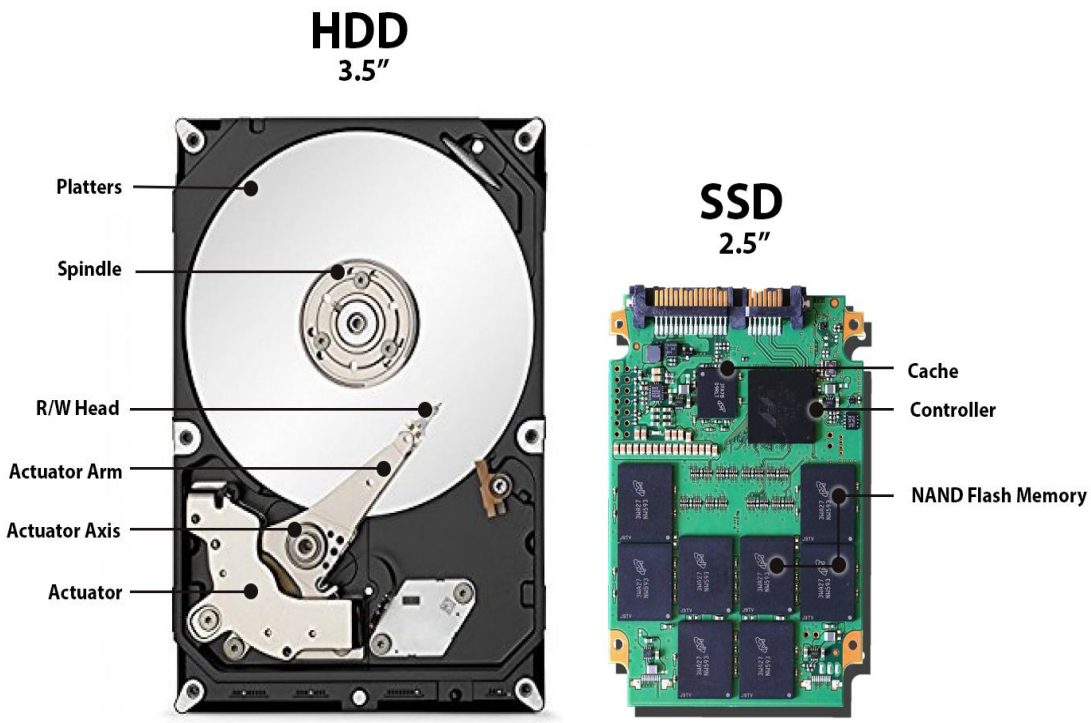
Disadvantage of SSD

- The cost of SSDs is far higher than that of traditional HDDs.
- Less capacity than HDD

SSD Vs HDD

Please watch the video in the URL below:

https://youtu.be/GN4YJaRA9Ss



HDD vs SSD

Comparison between different physical disk types

Item	SSD	SAS	NL-SAS	SATA
Performance	Very high	High	Relatively high	Relatively high
Reliability	Minor	High	Relatively high	Minor
Price	Very high	High	Relatively low	Low
Power consumption	Minor	High	Relatively low	Relatively low
Recommended scenario	High frequency immediate transactional data like database or bank transactions.	Suitable for users with discrete data access	Suitable for backup and data archiving (low data access)	Because of the high storage capacity they offer, SATA drives are commonly used for file sharing, email, web, backup, and archival data.